BILL OF MATERIALS & PROCUREMENT SCHEDULE (SHEET 1)

Cost Constraint: Budget Ceiling: INR 3,800.00 | **Total Reconciled Cost:** INR 3,264.00 | **Variance:** - INR 536.00 (Favorable)

1. CONSOLIDATED COMPONENT PROCUREMENT SCHEDULE

ITEM	REF	COMPONENT DESCRIPTION / PART NUMBER	MANUFACTURER / CHIPSET	QTY	UNIT COST (INR)	TOTAL COST (INR)	SUPPLIER
1	U1	Luckfox Pico Mini B (RV1103 SoC, 64MB DDR2, 128MB SPI NAND)	Rockchip / Luckfox	1	1,749.00	1,749.00	ThinkRobotics
2	U2	SC3336 3MP MIPI CSI-2 Camera Module (98.3 Deg FOV Lens)	SmartSens / Luckfox	1	800.00	800.00	Hubtronics
3	U3	USB 2.0 Wi-Fi Dongle (RTL8188EUS 802.11n 150Mbps SoftAP)	Realtek / Generic	1	180.00	180.00	Zbotic / Fabtolab
4	BT1	LiPo Rechargeable Battery 3.7V 1000mAh with PCM (JST-PH 2.0mm)	Probots (KP-503450)	1	340.00	340.00	Probots
5	U4	IP5306 Power Bank Module (2.1A Charge / 2.4A Boost / Load Sharing)	Injoinic / Generic	1	45.00	45.00	Techiesms
6	U5	LM2596 DC-DC Step-Down Buck Converter Module (12V to 5V DC)	TI / Generic	1	49.00	49.00	Robu.in
7	D1	TVS Diode SMBJ5.0A (600W Transient Voltage Suppressor, SMB)	Littelfuse / Generic	1	20.00	20.00	Robu.in
8	C1-C4	Ceramic Capacitors 100nF (104) 50V Radial (Pack of 10)	Murata / Generic	1 pk	15.00	15.00	Robu.in
9	R1-R5	Carbon / Metal Film Resistors Assorted (100R, 10k x2, 470R, 1k)	Generic (0.25W)	1 pk	20.00	20.00	Local Market
10	C5, C6	Aluminum Electrolytic Capacitors 470uF / 25V Radial	Nichicon / Generic	2	8.00	16.00	Robu.in
11	MISC	JST-PH 2.0mm 2-pin Connector Pairs and Interconnect Wires	Generic	1 set	30.00	30.00	Robu.in
				TOTAL RECONCILED PROCUREMENT COST:		**INR 3,264.00 (UNDER BUDGET)**	

ORGANIZATION / PROJECT: DASHCAM PROTOTYPE PROJECT	SIZE: A4	DOCUMENT NUMBER: ENG-DASH-BOM-001	DATE: 2026-10-08	SHEET: 1 OF 2	
DOCUMENT TITLE: BILL OF MATERIALS SPECIFICATION	REV: 1.4	DESIGN ENGINEER: Souvik H	ENGINEER REGISTRATION NO.: IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE: PROCUREMENT & COMPONENT ENGR	STATUS: RELEASED		

BILL OF MATERIALS & PROCUREMENT SCHEDULE (SHEET 2)

2. TECHNICAL COMPONENT ROLES & PROCUREMENT CHECKPOINTS

COMPONENT	CRITICAL TECHNICAL REQUIREMENT	VERIFICATION METHOD PRIOR TO ASSEMBLY
Luckfox Pico Mini B	Must feature 128MB SPI NAND onboard flash.	Confirm board boots from internal flash via USB MaskRom without MicroSD card present.
SC3336 3MP Module	Wide-angle optical lens (98.3 Deg FOV) with 15-pin FPC.	Verify CSI ribbon cable pitch (0.5mm) and pin 1 GND orientation with board socket.
USB Wi-Fi Dongle	Realtek RTL8188EUS chipset with SoftAP mode.	Verify device ID 0bda:8179 in Linux kernel; confirm '8188eu' driver binds automatically.
LiPo Battery Pack	3.7V 1000mAh with internal PCM protection board.	Verify open-circuit voltage (3.7V - 4.2V) and confirm JST-PH 2.0mm connector polarity (+/-).
IP5306 PMIC Module	Load-sharing capable synchronous boost converter.	Verify 5.0V output under 500mA load while simultaneously charging LiPo cell.
LM2596 Buck Regulator	Step-down automotive pre-regulator (3A peak).	Adjust trim potentiometer to precisely 5.00V DC before connecting downstream loads.
TVS Diode & Filter Network	SMBJ5.0A (600W) + 470uF bulk reservoir capacitors.	Verify TVS diode polarity (cathode towards +12V rail) and capacitor voltage ratings (25V min).

3. BUDGET VARIANCE & TRADE-OFF SUMMARY

- Initial Target Ceiling: INR 3,800.00 | Actual Reconciled BOM: INR 3,264.00 (Savings: INR 536.00).
- Eliminating the high-endurance MicroSD card saved ~INR 820.00 and prevented continuous vehicular write degradation.
- On-demand video recording is safely delegated to the client smartphone application (VLC) storage.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-BOM-001	DATE 2026-10-08	SHEET 2 OF 2	
DOCUMENT TITLE BILL OF MATERIALS SPECIFICATION	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE PROCUREMENT & COMPONENT ENGR	STATUS RELEASED		